kader Academy

Machine Safety Specialist (CE)

Module 3: Safety distances

TAB4: Risk reduction and standards
Tab4-3/9 20240818

kader Academy

1

Tab4-3/9-Sh.2

Safety distance standards

This module covers:
EN-ISO 13854:
Minimum gaps to avoid crushing of parts of the human body

EN-ISO 13855:
Positioning of safeguards with respect to the approach of the human body

EN-ISO 13857:
Safety distances to prevent hazard zones being reached by upper and lower limbs

2

Tab4-3/9-Sh.3

Minimum distances to prevent crushing

EN-ISO 13854: 2019
Safety of machinery - Minimum gaps to avoid crushing of parts of the human body

NEN-EN-ISO 13854
(en)

Veiligheid van machines - Minimumafstand ter voorkoming van het bekneld raken van menselijke lichaamsdelen (ISO 13854:2017,IDT)

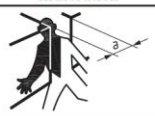
Safety of machinery - Minimum gaps to avoid crushing of parts of the human body (ISO 13854:2017,IDT)

Tab4-3/9-Sh.4

Minimum distances to prevent crushing

Table 1

Dimensions in millimetres

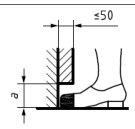
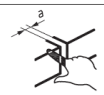
Part of body	Minimum gap *a*	Illustration
body	500	
head (least favourable position)	300	
leg	180	

Tab4-3/9-Sh.5

Minimum distances to prevent crushing

EN-ISO 13854

Table 1 *(continued)*

Part of body	Minimum gap *a*	Illustration
foot	120	
toes	50	
arm	120	
hand wrist fist	100	
finger	25	

5

Tab4-3/9-Sh.6

Safety distances for protective structures

EN-ISO 13857: 2008
Safety of machinery - Safety distances to prevent hazard zones being reached by upper and lower limbs

NEN-EN-ISO 13857
(en)

Veiligheid van machines - Veiligheidsafstanden ter voorkoming van het bereiken van gevaarlijke zones door bovenste en onderste ledematen (ISO 13857:2019,IDT)

6

Safety distances for protective structures

Contents:

- Safety distances for upper limbs
  - Reaching Overhead
  - Reaching over a shield
  - Reaching along a shield (with support)
  - Reaching through a shield
  - Reduction of safety distance through safety measures
- Safety distances for lower limbs
  - Reaching hazards by reaching through a protective structure

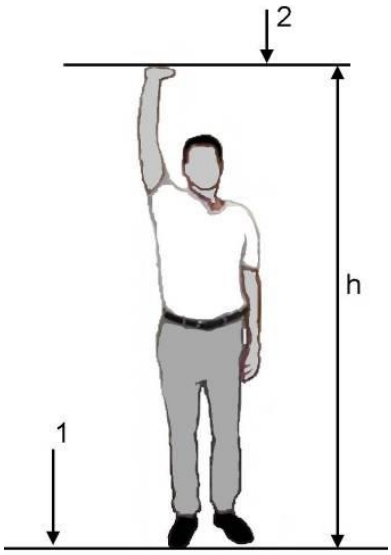
A number of applications are explained on the following sheets.

Safety distances protective structures

Reaching Overhead

- h > 2.7 m: no guarding (at low risk h>2.5)

1 = Reference level
2 = Danger zone



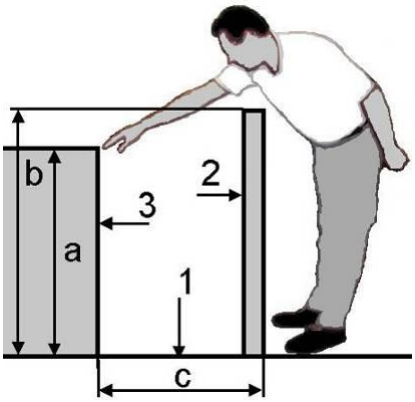
Tab4-3/9-Sh.9

Safety distances protective structures

Reaching over a protective structure

a = Danger Zone Height
b = Height of protective construction (guard)
c = Distance to danger zone

1 = Reference level
2 = Protective construction
3 = Danger zone



Tab4-3/9-Sh.10

Safety distances protective structures

Reaching over a protective structu

Table 2 — Reaching over protective structures

Dimensions in millimetres

h_h, height of the point of the hazard zone which is nearest to the area of upper limb reach[a]	h_{ps}, height of protective structure[b,c]									
	1 000	1 200	1 400	1 600	1 800	2 000	2 200	2 400	2 500	2 700
	s_h, horizontal safety distance of the point of the hazard zone which is nearest to the area of upper limb reach									
2 700	0	0	0	0	0	0	0	0	0	0
2 600	900	800	700	600	600	500	400	300	100	0
2 400	1 100	1 000	900	800	700	600	400	300	100	0
2 200	1 300	1 200	1 000	900	800	600	400	300	0	0
2 000	1 400	1 300	1 100	900	800	600	400	0	0	0
1 800	1 500	1 400	1 100	900	800	600	0	0	0	0
1 600	1 500	1 400	1 100	900	800	500	0	0	0	0
1 400	1 500	1 400	1 100	900	800	0	0	0	0	0
1 200	1 500	1 400	1 100	900	700	0	0	0	0	0
1 000	1 500	1 400	1 000	800	0	0	0	0	0	0
800	1 500	1 300	900	600	0	0	0	0	0	0
600	1 400	1 300	800	0	0	0	0	0	0	0
400	1 400	1 200	400	0	0	0	0	0	0	0
200	1 200	900	0	0	0	0	0	0	0	0
0	1 100	500	0	0	0	0	0	0	0	0

[a] For hazard zones above 2 700 mm, refer to 4.2.1.

[b] Protective structures less than 1 000 mm in height are not included because they do not sufficiently restrict movement of the body.

[c] Protective structures lower than 1 400 mm should not be used without additional protective measures.

Tab4-3/9-Sh.11

Safety distances protective structures

Reaching over a protective st
(only for a low risk)

Table 1 — Reaching over protective structures — Only minor injuries along with a low probability of occurrence

Dimensions in millimetres

h_h, height of the point of the hazard zone which is nearest to the area of upper limb reach	h_{ps}, height of protective structure[a]								
	1 000	1 200	1 400	1 600	1 800	2 000	2 200	2 400	2 500
	s_h, horizontal safety distance of the point of the hazard zone which is nearest to the area of upper limb reach								
2 500	0	0	0	0	0	0	0	0	0
2 400	100	100	100	100	100	100	100	100	0
2 200	600	600	500	500	400	350	250	0	0
2 000	1 100	900	700	600	500	350	0	0	0
1 800	1 100	1 000	900	900	600	0	0	0	0
1 600	1 300	1 000	900	900	500	0	0	0	0
1 400	1 300	1 000	900	800	100	0	0	0	0
1 200	1 400	1 000	900	500	0	0	0	0	0
1 000	1 400	1 000	900	300	0	0	0	0	0
800	1 300	900	600	0	0	0	0	0	0
600	1 200	500	0	0	0	0	0	0	0
400	1 200	300	0	0	0	0	0	0	0
200	1 100	200	0	0	0	0	0	0	0
0	1 100	200	0	0	0	0	0	0	0

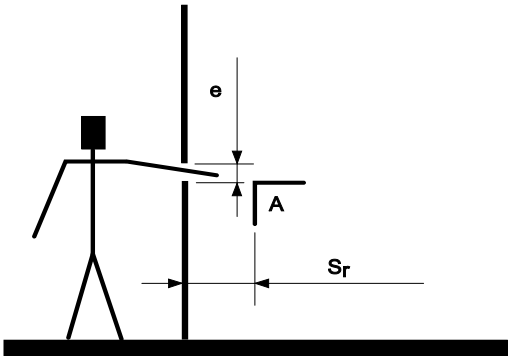
[a] Protective structures less than 1 000 mm in height are not included because they do not sufficiently restrict movement of the body.

Tab4-3/9-Sh.12

Safety distances protective structures

Reaching through a protective structure

e = size of opening in a protective structure
A = danger zone
Sr= safety distance



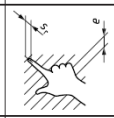
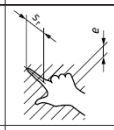
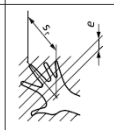
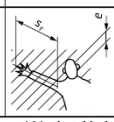
Tab4-3/9-Sh.13

Safety distances pr

Reaching through a protective struc

Table 4 — Reaching through regular openings — Persons of 14 years of age and above

Dimensions in millimetres

Part of body	Illustration	Opening	Safety distance to hazard zone, s_r		
			Slot	Square	Round
Fingertip		$e \leq 4$	≥2	≥2	≥2
		$4 < e \leq 6$	≥10	≥5	≥5
Finger up to knuckle joint		$6 < e \leq 8$	≥20	≥15	≥5
		$8 < e \leq 10$	≥80	≥25	≥20
Hand		$10 < e \leq 12$	≥100	≥80	≥80
		$12 < e \leq 20$	≥120	≥120	≥120
		$20 < e \leq 30$	≥850[a]	≥120	≥120
Arm up to junction with shoulder		$30 < e \leq 40$	≥850	≥200	≥120
		$40 < e \leq 120$	≥850	≥850	≥850

NOTE The bold lines within the table delineate that part of the body restricted by the opening size.

[a] If the length of the slot opening is ≤65 mm, the thumb will act as a stop and the safety distance may be reduced to ≥200 mm.

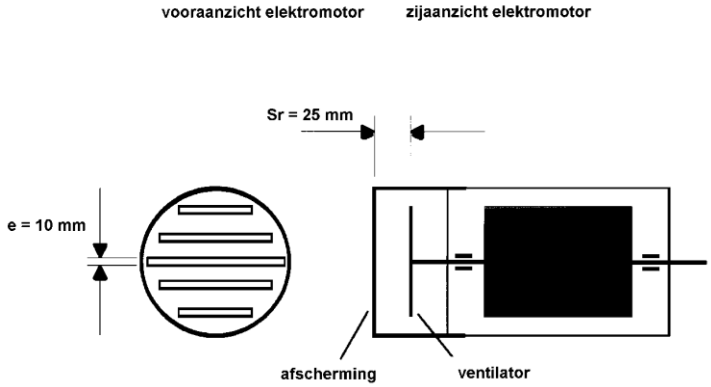
Tab4-3/9-Sh.14

Exercise 4-5

Reaching through a shield

See image and corresponding dimensions.

- Is the distance to the fan sufficient?
- If not, what are the possibilities to comply with the standard?



Safety distances for protective devices

EN-ISO 13855: 2010
Safety of machinery - Positioning of safeguards with respect to the approach of the human body

Nederlandse norm

NEN-EN-ISO 13855
(en)

Safety of machinery - Positioning of safeguards with respect to the approach speeds of parts of the human body (ISO 13855:2010,IDT)

Method (EN ISO 13855)

a. Identify the hazard and determine the risk (ISO 12100).
b. Is there a type C standard? Select the protection device from this and use the available formulas, if any.
c. Is there no type C standard? Select a suitable protective device using the ISO 13855 and calculate the minimum distance.
d. If it is possible to bypass the protection, an additional calculation must be carried out (Sec. 6.5, ISO 13855).
e. If combinations of protective devices are used, a minimum distance must be chosen, which takes into account every protection and the possibility of circumvention.
f. Calculate the minimum distance for each possibility to reach the danger zone. Then choose the most protective (largest) minimum distance.

Methode (EN ISO 13855)

g. If possible, the distances can be included in the design, otherwise go to step i.
h. Check that the placement of the security allows it to be undetected, otherwise go to step j.
i. Can parameters be adjusted or alternative safeguards used? If both are not possible, additional safety devices must be used.
j. Check that the position of the protective device allows people to stay between the protection and the danger zone without being detected. In this case, additional measures should be taken depending on the additional risk assessment.

Safety distance S

- S (mm)

  Minimum distance between the protective measure and the danger area.
  'If it takes 2 seconds to stop a machine, a light curtain must be placed 2 seconds away from danger'

Approach speed K

- K (mm/s)
  Approach speed parameter

  Usually K = 1600 mm/s
  Unless the distance between light curtain and danger area ≤ 500mm, then K= 2000 mm/s

Stopping time T

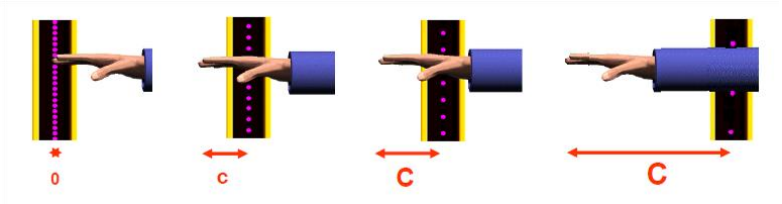
- T (s)
  'overall system stopping performance'. the time between the signalling of an intervention in the protected field up to and including the machine standstill

Depending, among other things, on:

- Control and security response times
- Mass inertia
- Drive force
- Power transfer
- Braking capacity
- Cycle times

Detection capability & Intrusion distance

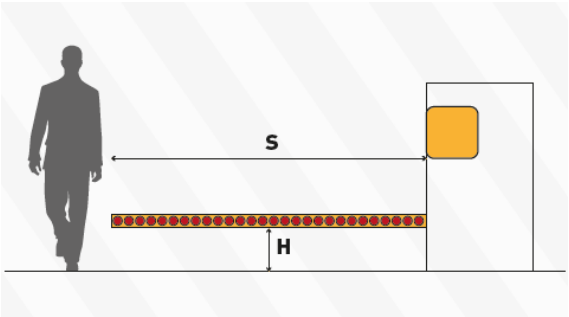
- d (mm)
  detection capability' of a security device is the possibility to detect a certain object size with certainty, i.e. the distance between 2 beams
- C (mm)
  Surcharge or 'Intrusion distance', is the distance that a part of the body has already traveled before it is detected.



Definitions

- S safety distance (mm)
- K approach speed (mm/s)
- T stop time(s)
- C surcharge (mm)
- D distance between two radii (mm)
- H height horizontally mounted light curtain (mm)

Safety zone



$$S = K \times T + C$$

$$C = (1.200 - 0{,}4 \times H)$$

$$\boxed{S = K \times T + (1.200 - 0{,}4 \times H)}$$

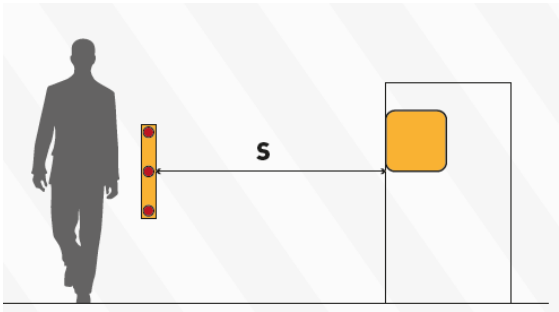
- There are additional requirements for height and resolution
- When applying a scanner, a distance Z (mm) is added. (correction of measurement errors, specified by the supplier of the scanner)

Beam height 'whole body access'

Number of beams	Heights above the reference plane, for example the floor mm
4	300, 600, 900, 1 200
3	300, 700, 1 100
2	400[a], 900

[a] For the lowest beam [see 6.2.2 a)] 400 mm can only be used when the risk assessment allows it.

Whole body access

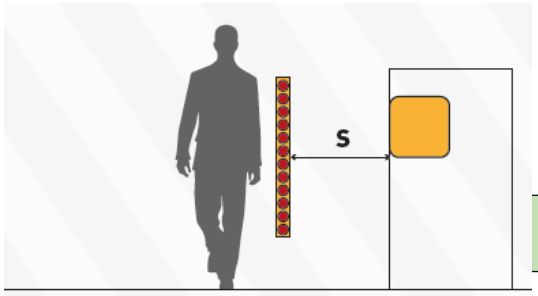


$$S = K \times T + C$$

$$C = 850mm$$

$$S = K \times T + 850$$

Reaching through detection



$$S = K \times T + C$$

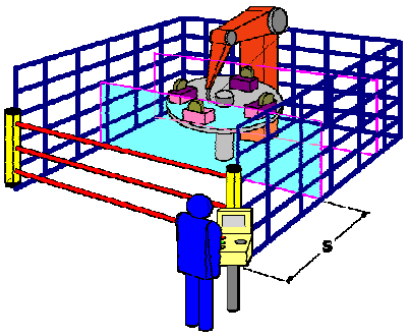
$$C = 8 \times (d - 14)$$

$$S = K \times T + 8 \times (d - 14)$$

- Er zijn aanvullende eisen aan minumumafstand en afstand tussen de stralen

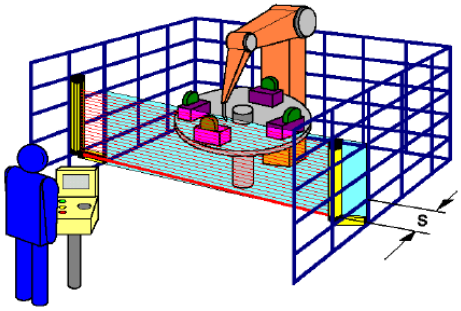
Exercise 4-6

- Machine stop time is 0.3 sec.
- Determine the minimum safety distance.

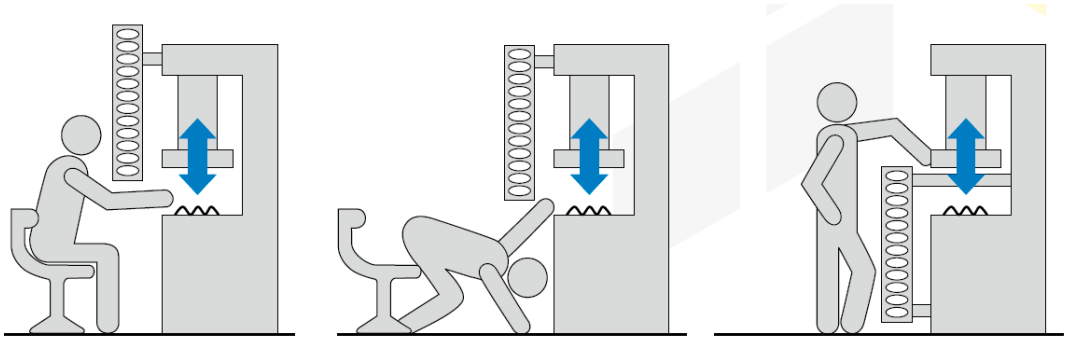


Exercise 4-7

- Machine stop time is 0.3 s.
- Resolving power 14mm.
- Determine the minimum safety distance.



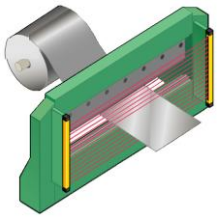
Reaching under and reaching over

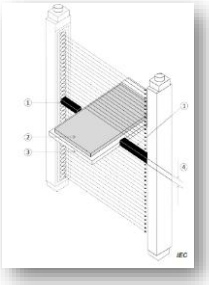


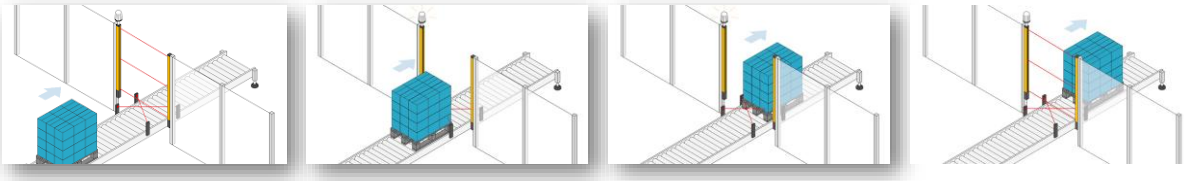
ondergrijpen *ondergrijpen vanuit lagere positie* *overgrijpen*

Muting

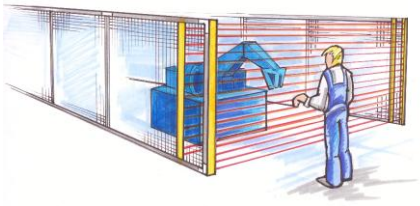
EN IEC 62046 provides explanations and examples of:
Muting for people access
Muting for product access
Mute dependent override
Fixed blanking
Floating blanking
Reduced resolution





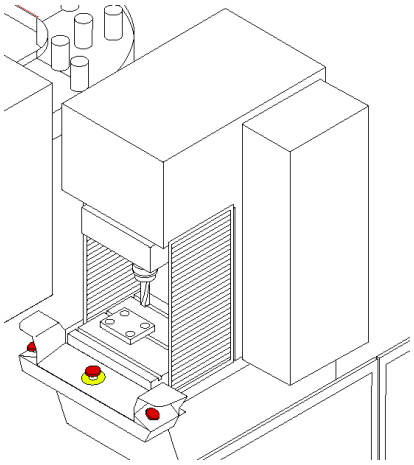


Light curtains



Also think of shooting loose objects, programmed range, etc

Two-hand control



Two-hand control

The choice of this device should be considered with care as it generally protects ONLY the person carrying out the operation and does not prevent others who are nearby from accessing the danger areas.

EN-ISO 12100

Two-hand control

- Within range moving parts not in operation
- Moving parts cannot be reached
- Distance between organs > 260 mm
- Simultaneous operation
- Stopping when letting go
- New start only after new starting position
- Defect = downtime
- Redundant and cyclically controlled

EN ISO 13851

Two hand control

Minimale afstand S van de dichtstbijzijnde drukknop naar de gevarenzone :

$S = K \times T + C$

- K = 1600 mm/s
- C = 250 mm

$S = 1600 \times T + 250$

Interlocking guards with guard locking

Minimum distance S from the nearest opening to the danger zone :

$$S = K \times T + C$$

- K = 1600 mm/s
- C is the safety distance from Table 4 or 5 of EN ISO 13857 if fingers or a hand can pass through the shield
- T may be reduced by the time it takes to open the gate far enough to reach the hazard. Use table 4 or 5 of EN ISO 13857 for this as well
- If it is not possible to achieve the minimum distance, locking can be applied (see EN ISO 14119)

kader
Academy